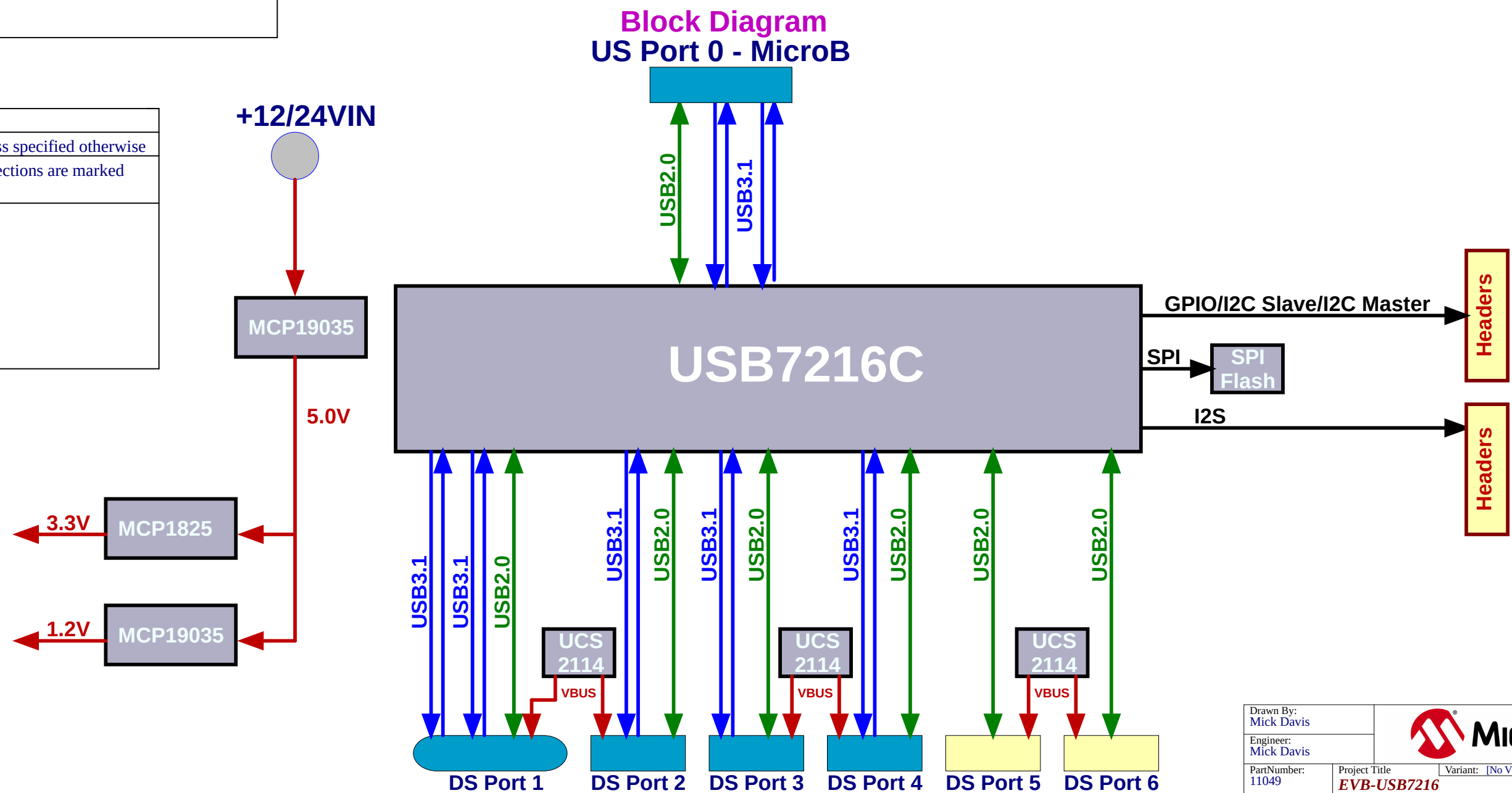


EVB-USB7216

Table of Contents	
Sheet	Description
1	EVB-USB7216
2	USB7216 Part A
3	USB7216 Part B
4	Memory
5	USB MicroB UFP
6	USB3.1 (Type-C) DFP1
7	USB3.1(Type-A) DFP2-4
8	USB2.0 (Type-A) DFP5-6
9	Voltage Regulators

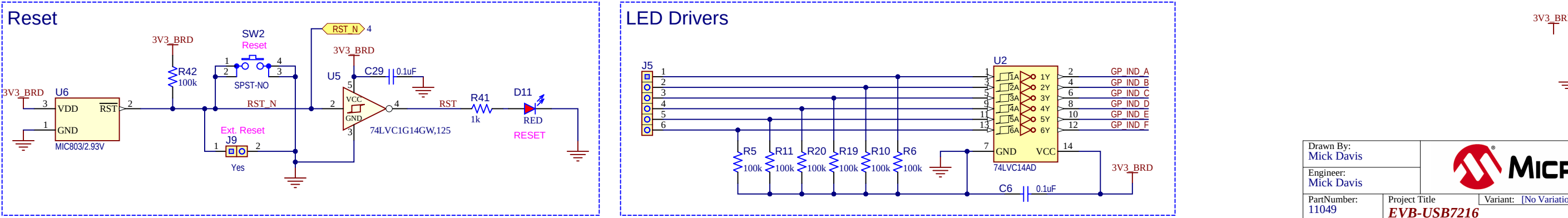
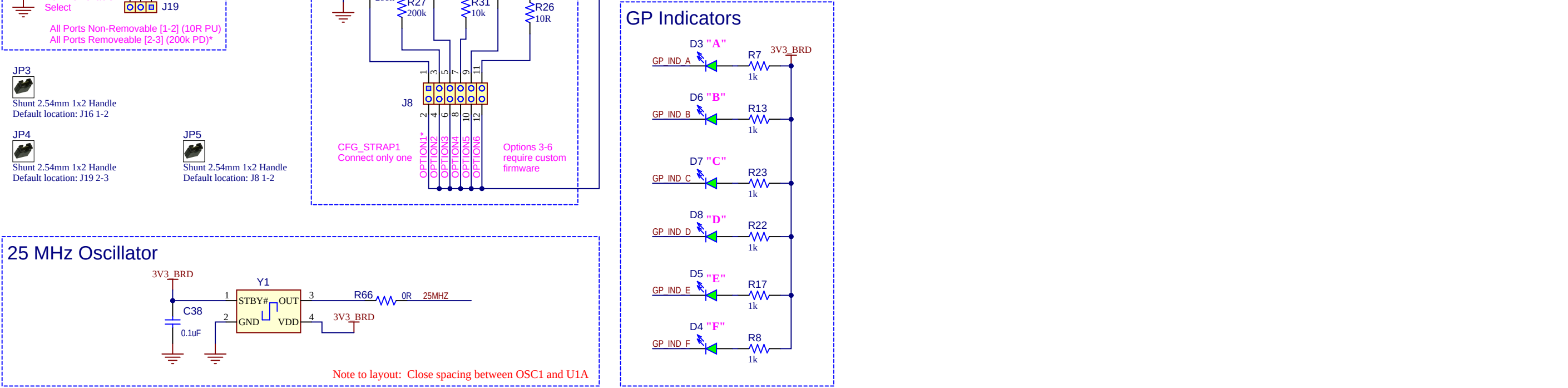
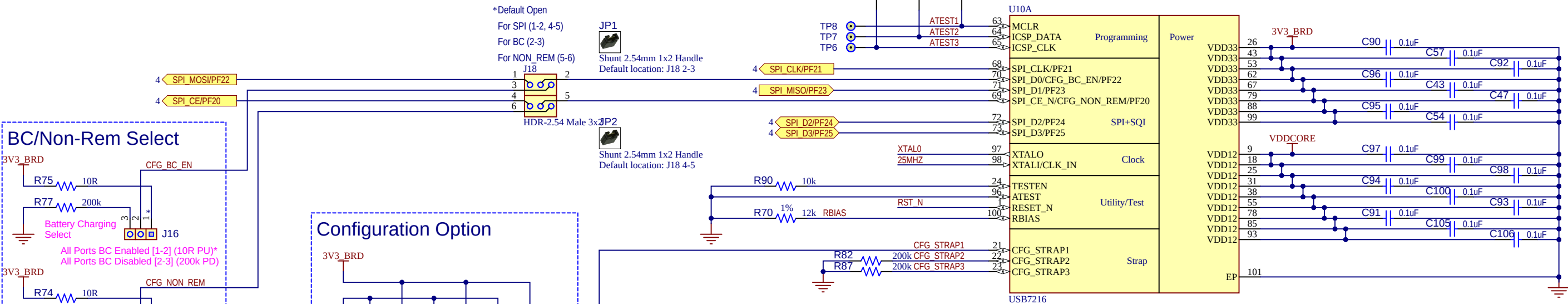
Revision History			
Revision	Date	Revision Summary	Author
1.0	10/02/2019	DevTools Transfer	M. Davis
2.0	8/12/2020	Full Revision	M. Davis

Notes	
1	All resistors are 1% unless specified otherwise
2	Shunt jumper default selections are marked with an asterisk [*].



Drawn By: Mick Davis		 MICROCHIP			
Engineer: Mick Davis					
PartNumber: 11049	Project Title EVB-USB7216	Variant: [No Variations]			
Sheet Title EVB-USB7216		Designed with  Altium.com			
Size	SCH #: 03-11049			Rev:2.0	Date: 10/22/2025
Tableld	PCB #: 04-11049			Rev:2.0	Sheet 1 of 9
File: 11049-TOC SchDoc					

USB7216 Part A



USB7216 Part B

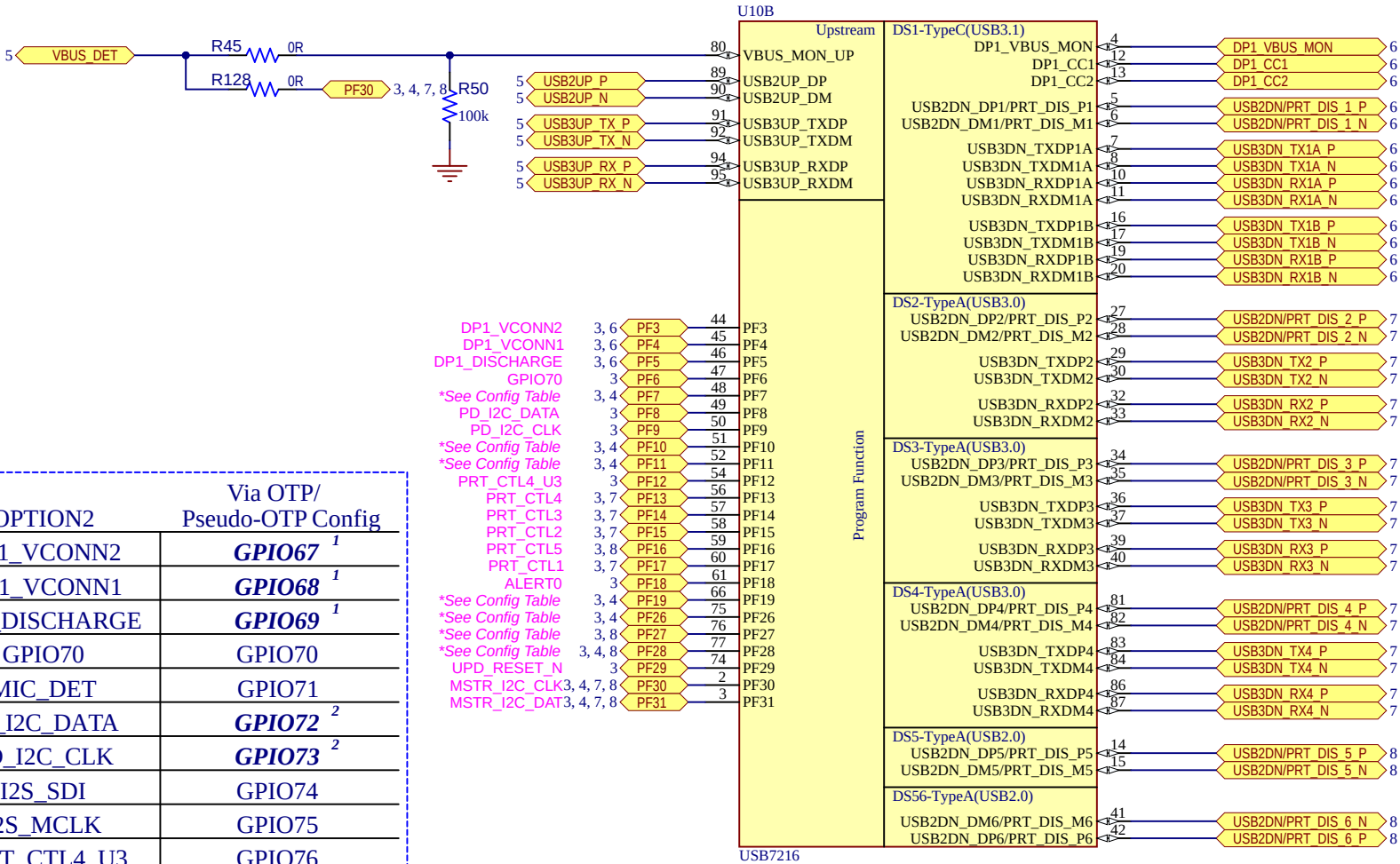
Default PF Functions by Configuration Option

	OPTION1	OPTION2	Via OTP/ Pseudo-OTP Config
PF3	DP1_VCONN2	DP1_VCONN2	GPIO67 ¹
PF4	DP1_VCONN1	DP1_VCONN1	GPIO68 ¹
PF5	DP1_DISCHARGE	DP1_DISCHARGE	GPIO69 ¹
PF6	GPIO70	GPIO70	GPIO70
PF7	GPIO71	MIC_DET	GPIO71
PF8	PD_I2C_DATA	PD_I2C_DATA	GPIO72 ²
PF9	PD_I2C_CLK	PD_I2C_CLK	GPIO73 ²
PF10	PRT_CTL2_U3	I2S_SDI	GPIO74
PF11	PRT_CTL3_U3	I2S_MCLK	GPIO75
PF12	PRT_CTL4_U3	PRT_CTL4_U3	GPIO76
PF13	PRT_CTL4	PRT_CTL4	GPIO77
PF14	PRT_CTL3	PRT_CTL3	GPIO78
PF15	PRT_CTL2	PRT_CTL2	GPIO79
PF16	PRT_CTL5	PRT_CTL5	GPIO80
PF17	PRT_CTL1	PRT_CTL1	GPIO81
PF18	ALERT0	ALERT0	GPIO82 ²
PF19		I2S_SDO	GPIO83
PF20	SPI_CE_N	SPI_CE_N	GPIO84 ³
PF21	SPI_CLK	SPI_CLK	GPIO85 ³
PF22	SPI_D0	SPI_D0	GPIO86 ³
PF23	SPI_D1	SPI_D1	GPIO87 ³
PF24	SPI_D2	SPI_D2	GPIO88 ³
PF25	SPI_D3	SPI_D3	GPIO89 ³
PF26	SLV_I2C_CLK	I2S_SCK	GPIO90
PF27	SLV_I2C_DATA	PRT_CTL6	GPIO91
PF28	PRT_CTL6	I2S_LRCK	GPIO92
PF29	UPD_RESET_N	UPD_RESET_N	GPIO93 ²
PF30	MSTR_I2C_CLK	MSTR_I2C_CLK	GPIO94
PF31	MSTR_I2C_DATA	MSTR_I2C_DATA	GPIO95

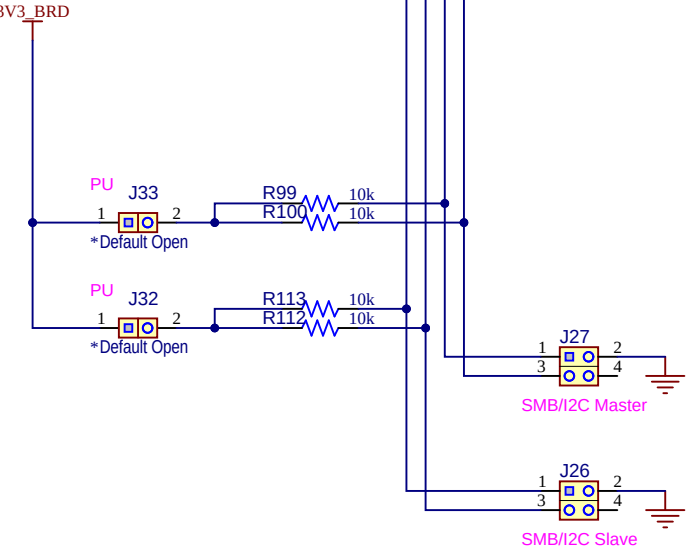
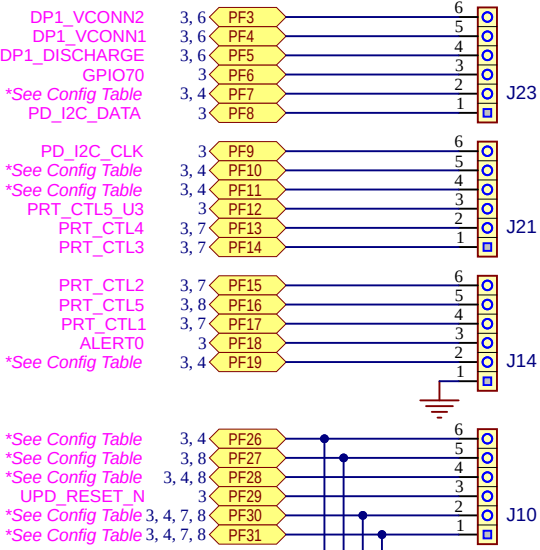
¹ These pins may not be used as GPIOs on this design. They may be used only if Downstream Port 1 is connected to a legacy port(Type-A)

² These pins may not be used as GPIOs on this design. They may only be used in special applications where USB Power Delivery is not implemented

³ These pins may not be used as GPIOs on this design. Can be used only in special applications where SPI Flash is not utilized



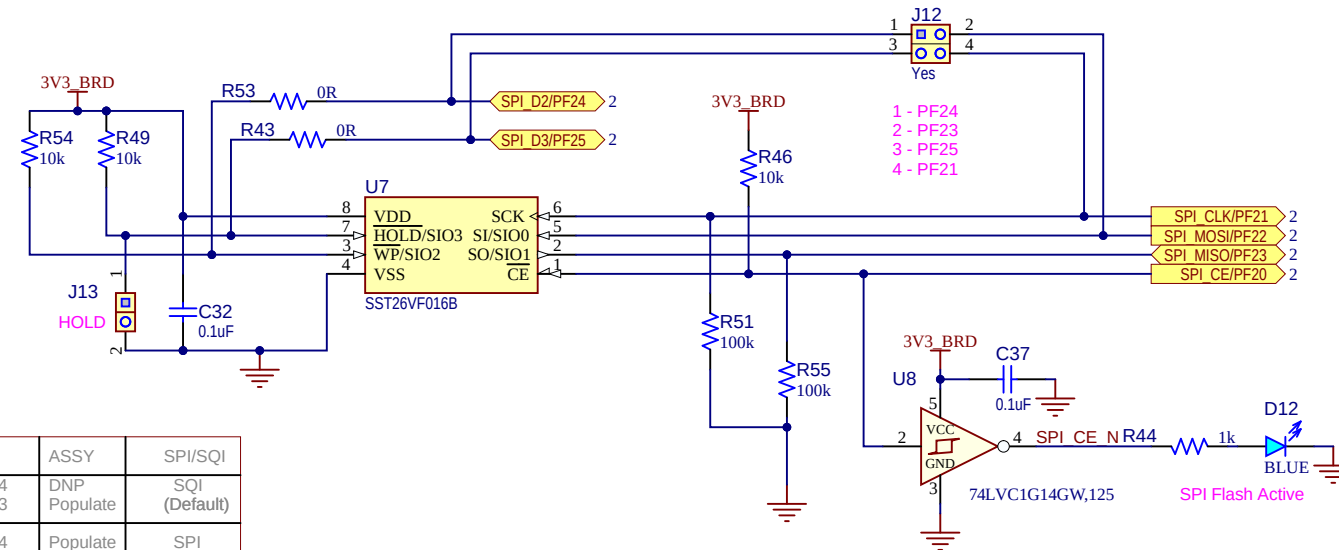
GPIO and SMBUS options



Memory

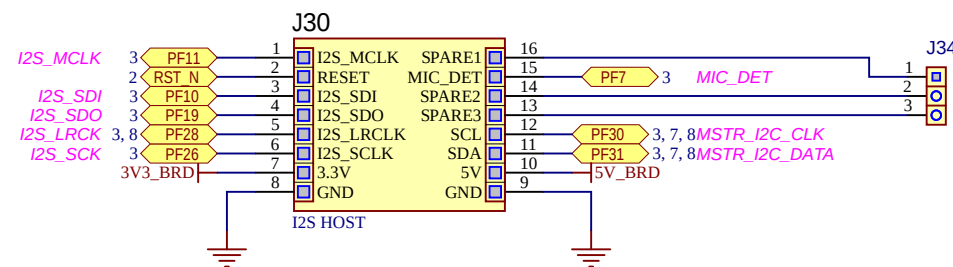
SPI Flash

Layout: Arrange all signals in this block with minimal stubs



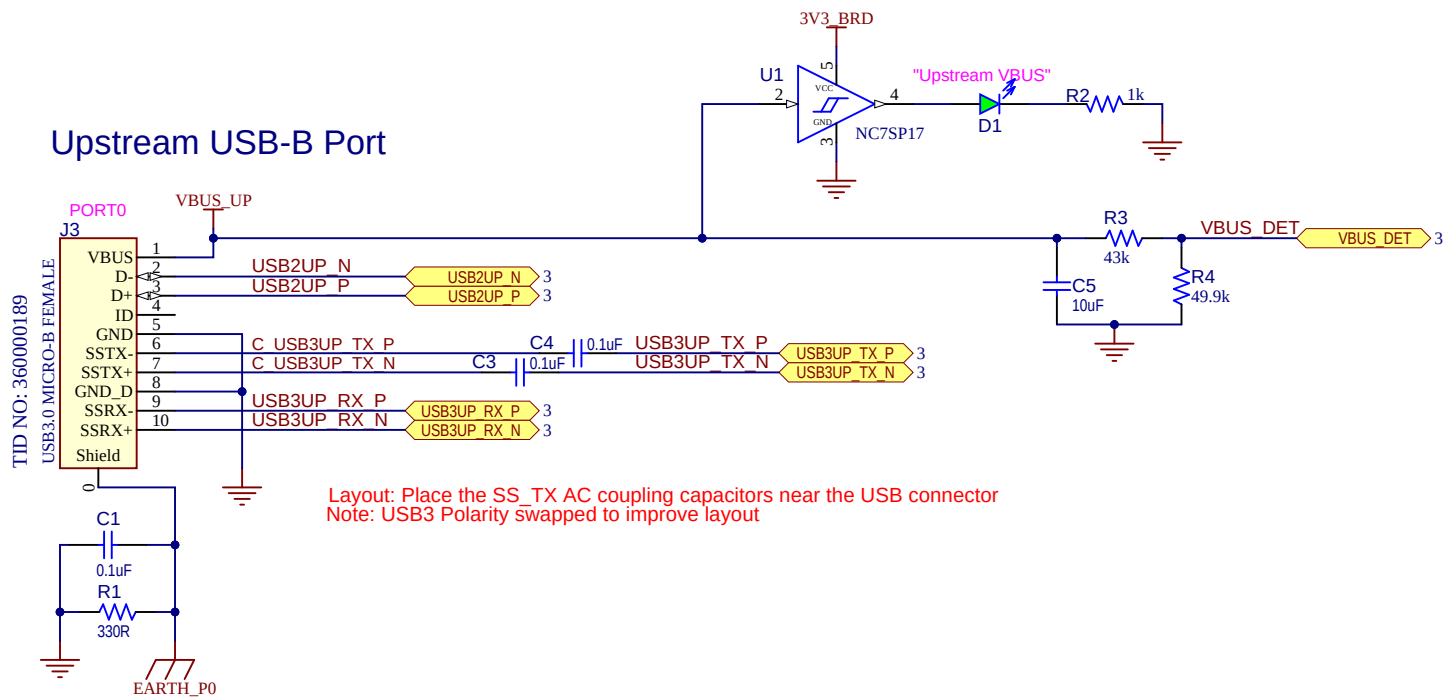
Resistor	ASSY	SPI/SQI
R49 & R54 R43 & R53	DNP Populate	SQI (Default)
R49 & R54 R43 & R53	Populate DNP	SPI

I2S Codec

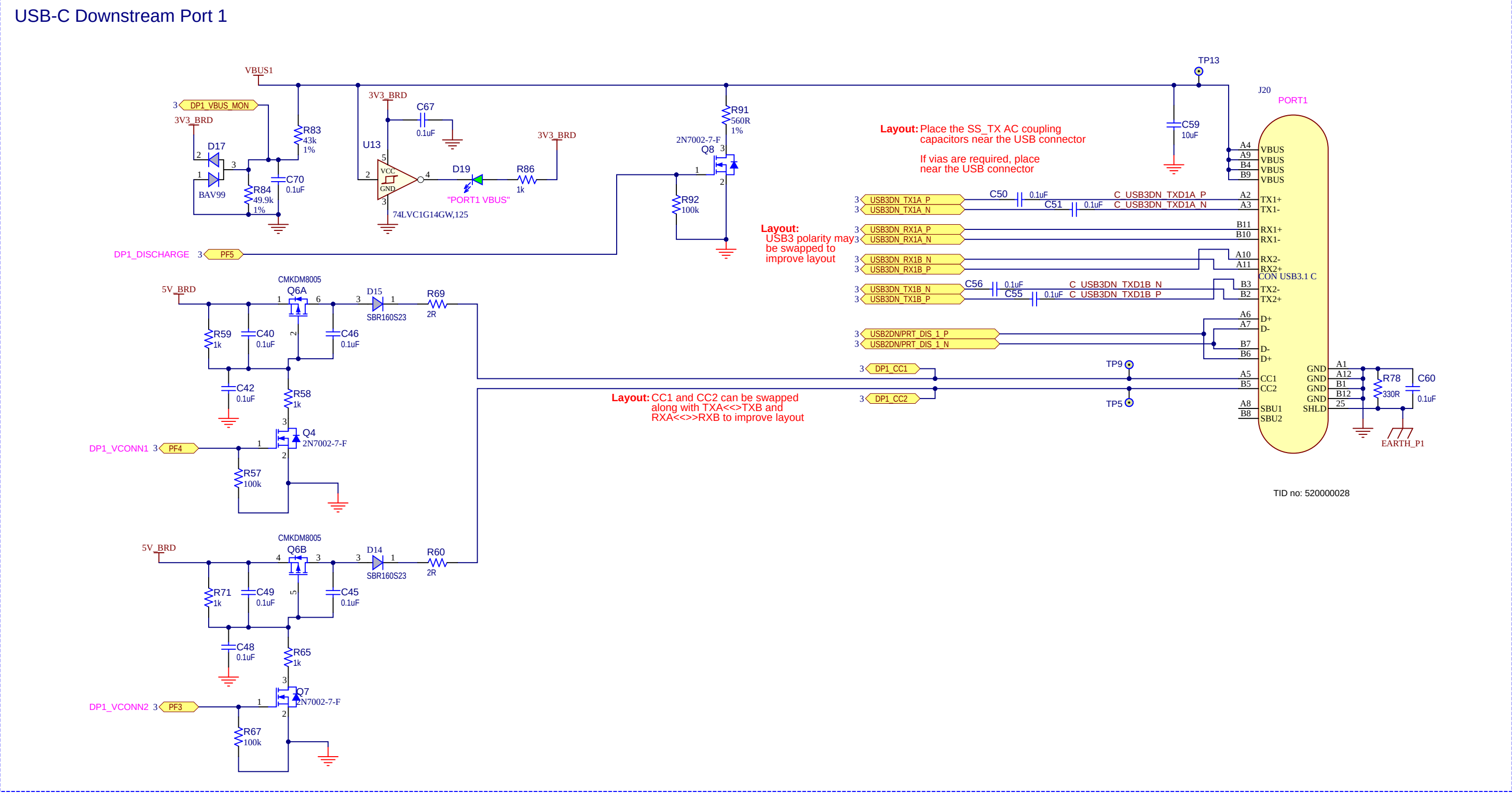


Drawn By: Mick Davis		 MICROCHIP			
Engineer: Mick Davis					
PartNumber: 11049	Project Title EVB-USB7216	Variant: [No Variations]			
Sheet Title Memory		Designed with  Altium.com			
Size Tabloid	SCH #: 03-11049 PCB #: 04-11049			Rev:2.0 Rev:2.0	Date: 10/22/2025 Sheet 4 of 9
File: 11049-Memory_SchDoc					

USB Type-B Upstream Port



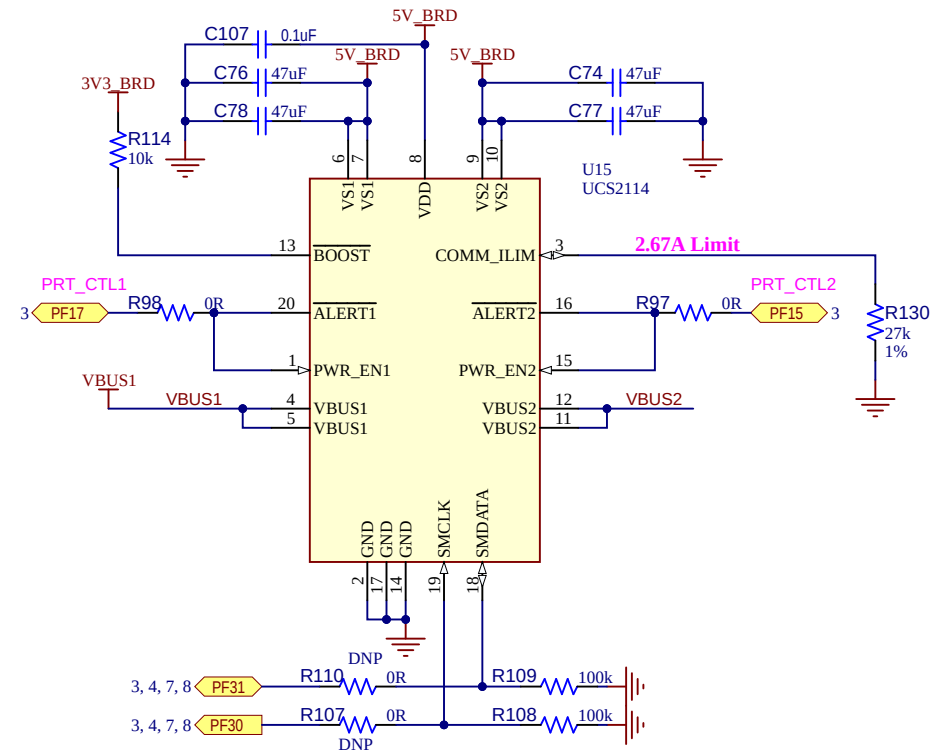
USB3.1 (Type-C) Downstream Port 1



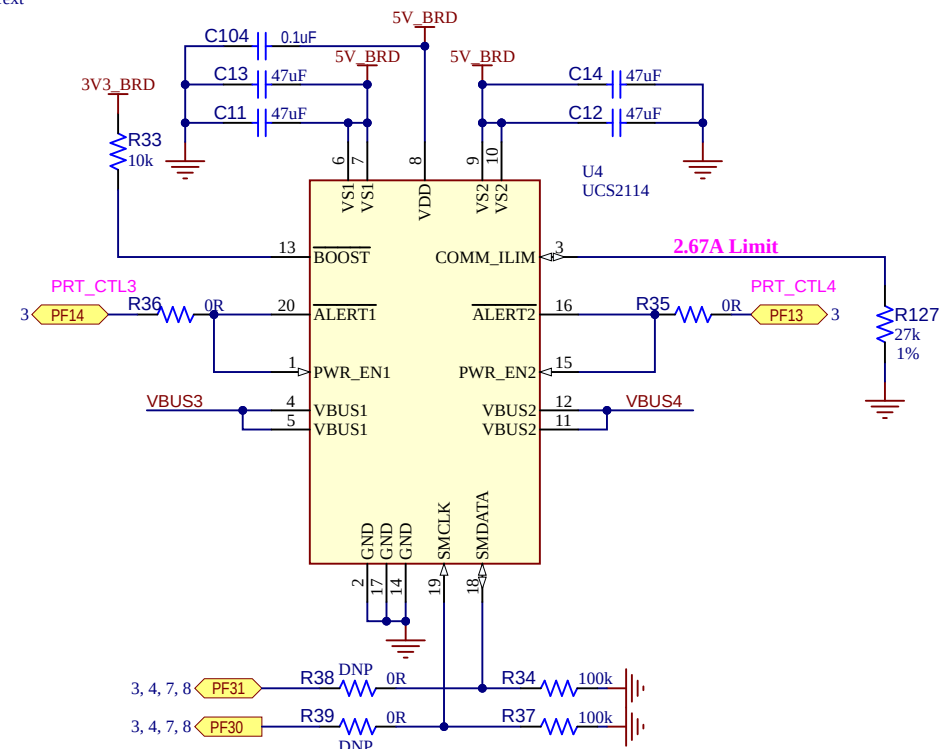
Drawn By: Mick Davis			
Engineer: Mick Davis			
PartNumber: 11049	Project Title EVB-USB7216	Variant: [No Variations]	
Sheet Title USB 3.1 (Type-C) Downstream Port 1			
Size Tabloid	SCH #: 03-11049 PCB #: 04-11049	Rev: 2.0 Rev: 2.0	Date: 10/22/2025 Sheet 6 of 9
File: 11049-USB-C Downstream 1.SchDoc			Designed with  Altium.com

USB3.1 (Type-A) Downstream Ports 2-4

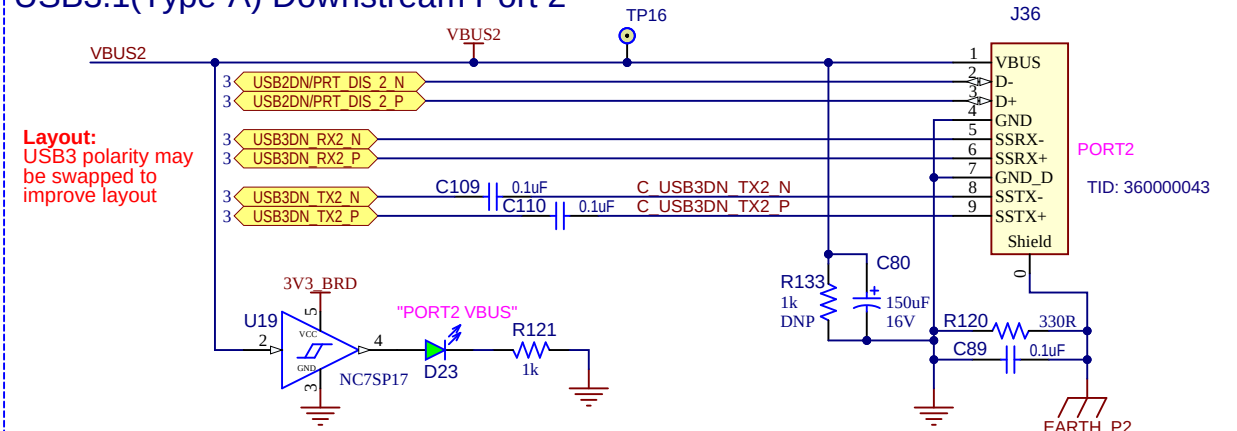
UCS2114 Power Switch For Port1 and Port2



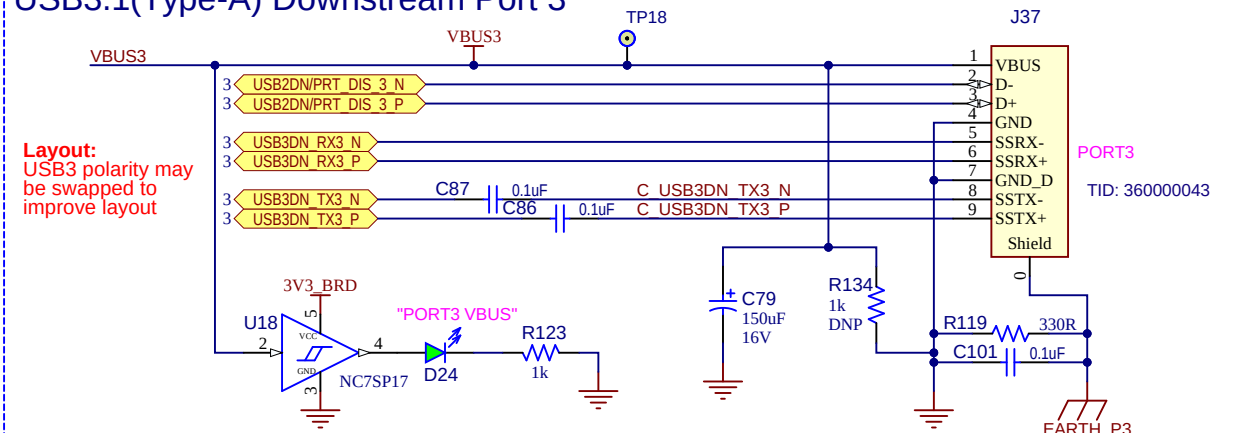
UCS2114 Power Switch For Port3 and Port4



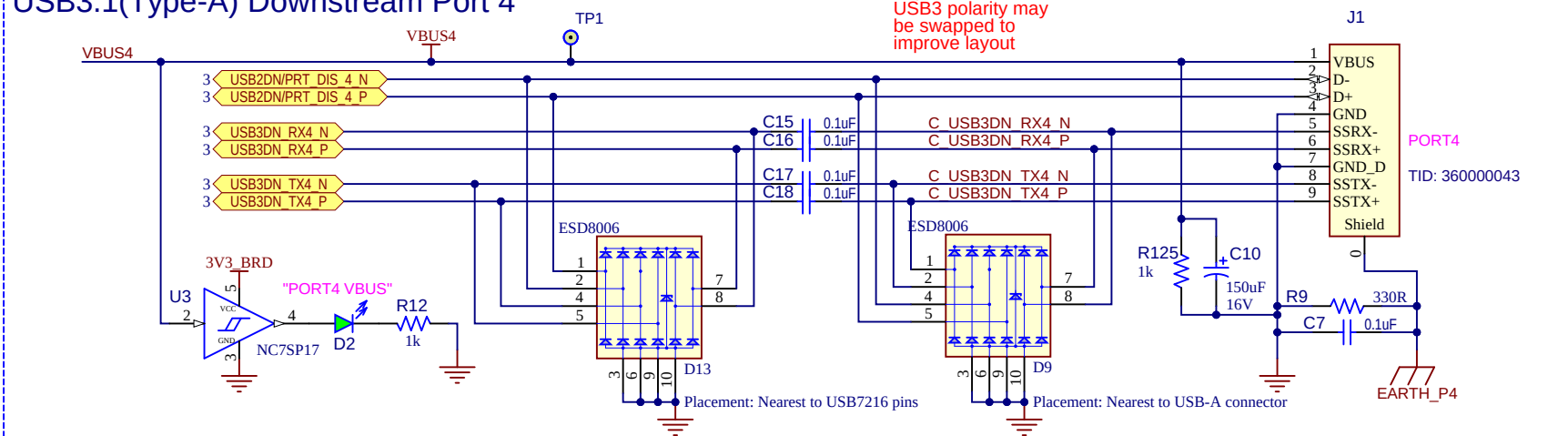
USB3.1(Type-A) Downstream Port 2



USB3.1(Type-A) Downstream Port 3



USB3.1(Type-A) Downstream Port 4



Drawn By: Mick Davis
Engineer: Mick Davis



PartNumber: 11049	Project Title EVB-USB7216	Variant: [No Variations]
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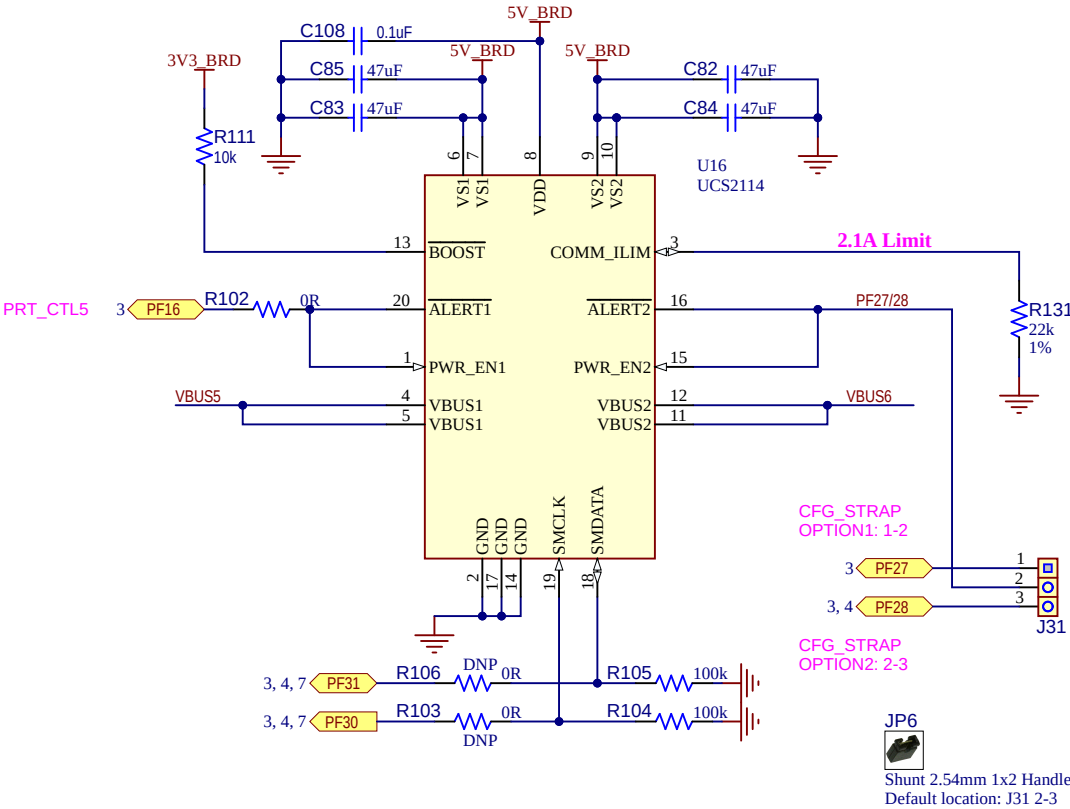
Sheet Title	USB3.1 (Type-A) Downstream Ports 2-4
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Size	SCH #: 03-11049	Rev: 2.0	Date: 10/22/2025
Tabloid	PCB #: 04-11049	Rev: 2.0	Sheet 7 of 9
File: 11049_USB_A_Downstream_2.4_SchDoc			

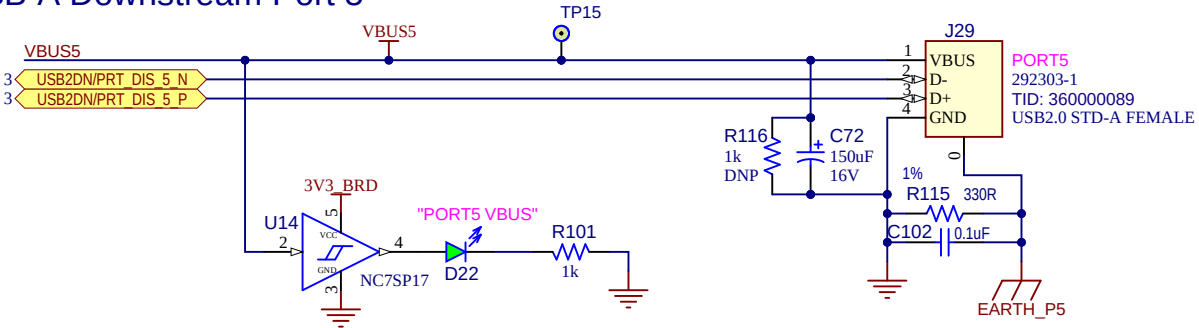
Designed with
Altium
Altium.com

USB2.0 (Type-A) Downstream Ports 5-6

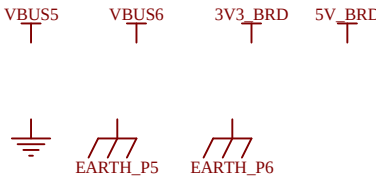
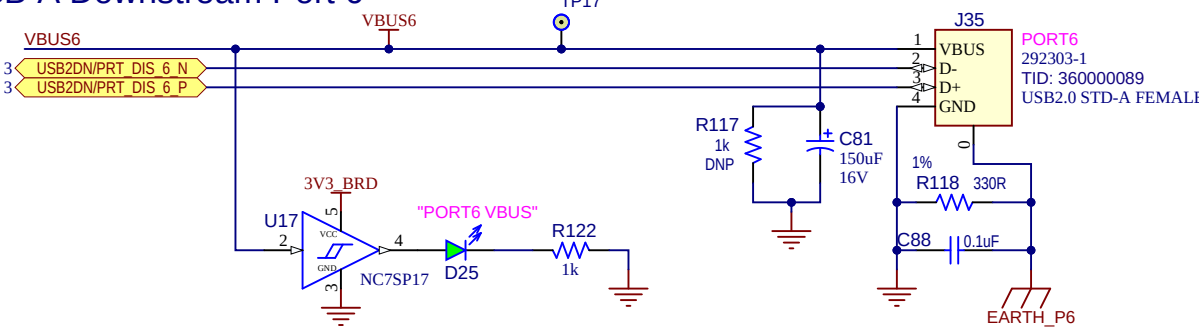
UCS2114 Power Switch For Port5 and Port6



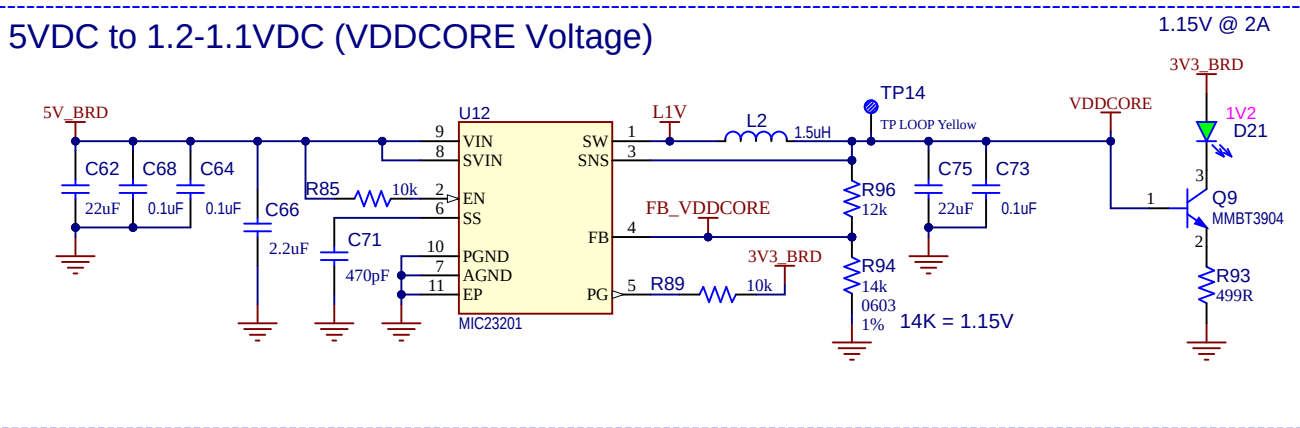
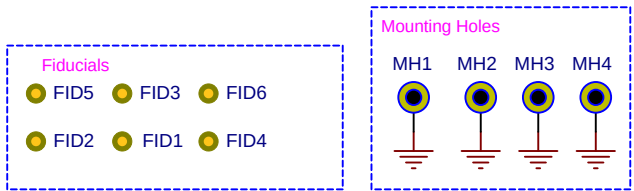
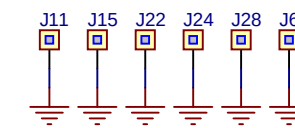
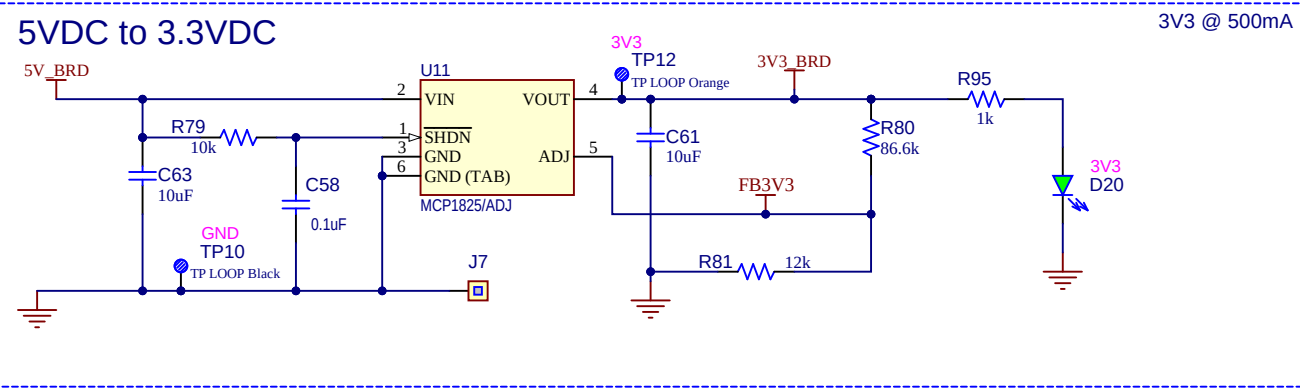
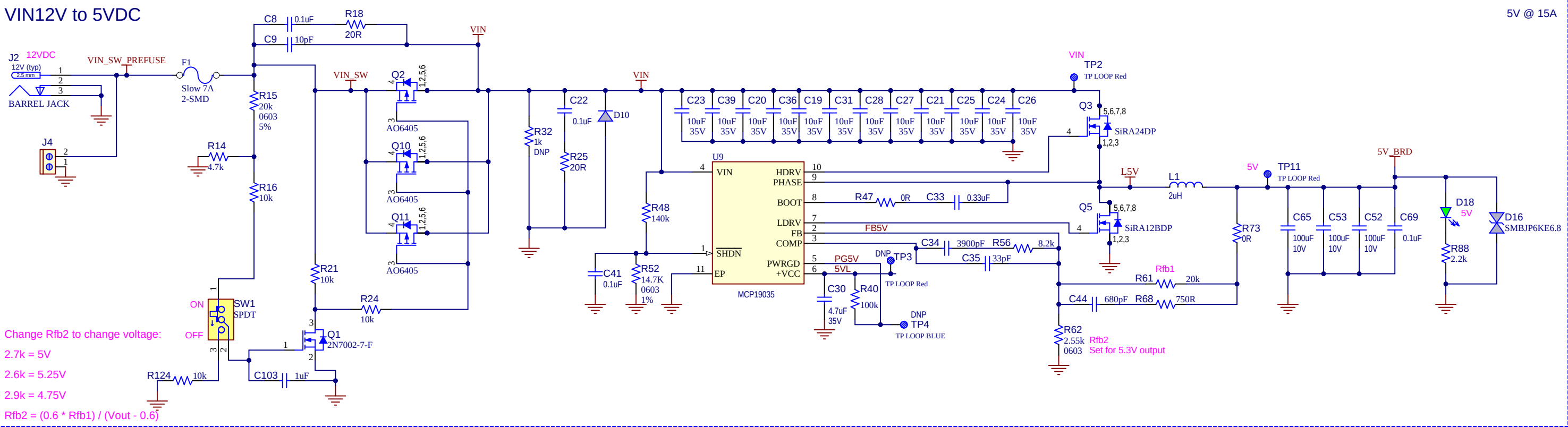
USB-A Downstream Port 5



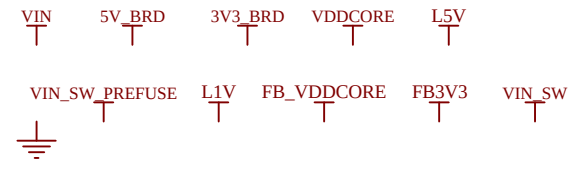
USB-A Downstream Port 6



Voltage Regulators

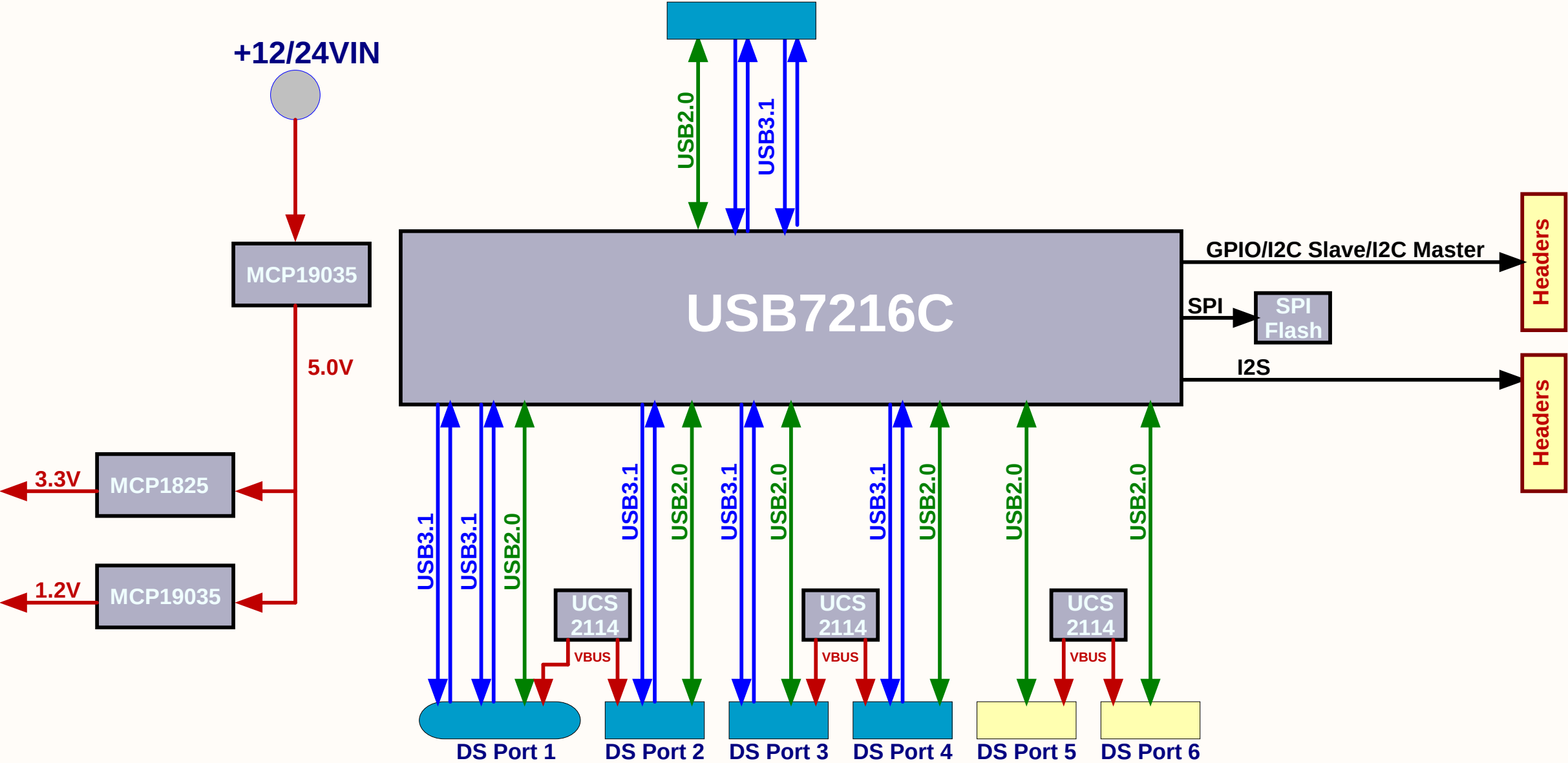


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Drawn By: Mick Davis			
Engineer: Mick Davis			
PartNumber: 11049	Project Title EVB-USB7216	Variant: [No Variations]	
Sheet Title Voltage Regulators		Designed with 	
Size Tabloid	SCH #: 03-11049 PCB #: 04-11049	Rev:2.0 Rev:2.0	Date: 10/22/2025 Sheet 9 of 9
File: 11049-Voltage_Regulators.SchDoc			

Block Diagram
US Port 0 - MicroB



Title		
Size B	Number	Revision
Date:	10/22/2025	Sheet of
File:	Sheet1.SchDoc	Drawn By: